AI USAGE REFLECTION & LOG | CHAOS ENGINE LABS GROWTH CASE

Deliverable 6 | Author: Head of Growth | As of Date: September 30, 2025

1. Two Critical Moments When AI Tools Gave Erroneous Output & How We Fixed Them

MISTAKE 1: Censored-Median Contamination & Single-Day Payer Window in Data Cleaning

• The Error: An initial AI code pass cleaned the July 14-16 ad glitch by computing baseline ad rates across all non-overlapping cohorts without filtering for cohort maturity (age_days >= h). Because immature cohorts hold frozen zero increments, the median normal rate collapsed to 0.0 at d60, d90, d120, and d180, wiping out 99% of legitimate mature ad revenue for valid cohorts overlapping mid-July. Simultaneously, the script divided multi-day incremental IAP (d7->d14 is 7 days, d14->d30 is 16 days) by single-day active payer counts (payers_d{h}), falsely flagging 142 legitimate US repeat-purchase cohorts on Net-B and Net-D as corrupted anomalies.
• Detection: We cross-reconciled cleaned cohort revenue totals against telemetry_daily.csv mediation totals ($299,079 ad rev in Game A) and audited row-level payer spend histories on US lines.
• Deterministic Fix: We enforced age_days >= h on baseline rate calculations, implemented install-weighted cell rates, and restricted IAP anomaly cleaning strictly to confirmed .xx00 SKU multiples (e.g. $469.43, $938.86) exceeding $150 with zero or near-zero window payers.

MISTAKE 2: Unconditioned Partial-Age Predictions on P7/P8 & Simpson's Paradox on Net-A Non-US

• The Error: The first LTV modeling pass masked partial d60 revenues for August 4-17 cohorts (ages 44-57 days on Sep 30), predicting Day-60 ROAS of 0.9228 for P7 and 0.7409 for P8. However, by September 30, those cohorts had ALREADY BANKED 0.9306 (P7) and 0.7734 (P8) in realized cash—predicting less money than was already in the bank! Simultaneously, when projecting Net-A Non-US (P2), the AI divided all-time Jan-Jun LTV ($1.15, dominated by expensive Tier-1 tests) by September's cheaper country-mix CPI ($0.65), creating a Simpson's Paradox denominator mismatch that artificially inflated projected ROAS to 1.78x. Furthermore, it bootstrapped parameter means rather than residual quantiles, yielding an uncalibrated 27.5% CI coverage.
• Detection: We ran strict monotonicity assertions (Obs_Prev < Banked_Sep30 < low_80 < prediction < high_80) and held-out coverage backtests.
• Deterministic Fix: We reformulated P1-P8 to condition directly on realized revenue banked as of September 30, projecting only unelapsed days via power-law elapsed fractions. We projected ROAS in consistent cohort-spend space and calibrated conformal residual quantiles (~79% empirical coverage).

2. Where AI Saved Significant Time

• Vectorized 8-Problem Forensic Audit: Rapidly joining and reconciling 166,000+ rows across 12 raw CSVs to verify $392,075.67 Game A spend and $669,975.34 Game B spend to 100.00% precision.
• Hierarchical Dual-Stream Survival Modeling: Accelerating the formulation of decoupled power-law retention and shifted lognormal IAP monetization with Empirical Bayes shrinkage across 62 market cells.
• Constrained Non-Linear KKT Spend Optimization: Solving the 40-cell non-linear constrained optimization across the CFO-vs-CEO Pareto frontier in minutes.

3. Short List of What Was Intentionally Skipped and Why

• User-Level Deep Learning Models: Raw data is cohort-aggregated; complex neural networks overfit small-sample cells and lack interpretability. Parametric survival models + Empirical Bayes provide full transparency, strict monotonicity, and verifiable credible intervals.
• Individual Budget Optimization for Micro-Territories (<$10/wk Spend): Net-A contains over 80 long-tail countries with volatile 1-5 install volumes. We grouped these into 4 economically homogeneous regional tiers to preserve statistical power while documenting all constituent country codes.